

14: Use of the Microscope

Learning Objectives

- Identify the parts of the microscope and their functions.
- Become familiar with the 3 variations of light microscopy.
- Learn how to use the microscope effectively, and particularly the oil immersion lens.
- Prepare wet mounts and stained smears of microbial suspension.

Wet Mounts and Stained Smears

The microscope is absolutely essential to the microbiology lab: most microorganisms cannot be seen without the aid of a microscope, save some fungi. And, of course, there are some microbes which cannot be seen even with a microscope, unless it is an electron microscope, such as the viruses.

You will be using an assigned light microscope for a variety of lab exercises through the semester, everything from viewing pond water to identification of your unknown bacterium. Therefore, it is extremely important that you understand how to use the microscope effectively and how to use different types of microscopy----**brightfield**, **phase-contrast**, and **darkfield**. **Phase-contrast and darkfield microscopies are used for wet mounts, whereas brightfield can be used for both wet mounts and stained specimens.**

You will also get your first exposure to the preparation of a bacterial smear and subsequent staining of it. However, you are making a simple stain, using only one dye. Everything on the slide will be the same color, but you can distinguish among shapes, sizes, and arrangements of bacteria.

MATERIALS NEEDED

- microscope
- pond water
- microscope slides
- cover slips
- oil dropper
- prepared slides of **Bacillus**
- toothpicks
- dye kits

THE PROCEDURES

Take a look at the sections outlined below on microscope use.

Prepare a wet mount of pond water

1. Go **DOWN** into the algae and muck to get a really good sample of protozoa and algae.
2. Focus on the sample using 10X, then go to 40X (**NOT** 100X). Start with brightfield, then switch over to darkfield and phase-contrast (see directions below).
3. Practice with the microscope, changing condensers settings, using different lenses. It is **NOT** important to identify protozoa or algae.

Prepare a smear and simple stain of the material between your teeth.

1. Take a sterile toothpick, remove some solid material between your wisdom teeth, and mix it into a drop of water so that you have a suspension spread over the middle third of the microscope slide. **NO COVER SLIP USED!**
2. Let the slide **air-dry**.
3. Heat-fix the dry smear by running the slide quickly through the flame a few times. If your fingers get hot, you have heat-fixed **TOO MUCH**.
4. Place the slide on the wire over the stain tray. Flood the smear with crystal violet: let sit for 1 minute.
5. Wash the slide **WELL** with distilled water. Blot the smear slide with your bibulous paper pad.
6. Focus on the sample using the 10X lens(**Be SURE that you are on brightfield microscopy**): you should see masses of purple material, most of it too small to see.

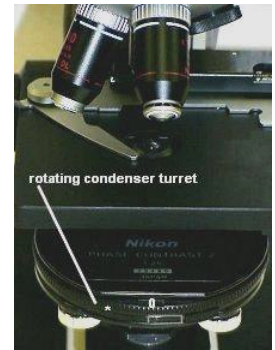
7. Ready to go to 100X now? **Do NOT** move the stage or the adjustments knobs before you **follow the directions below**.
8. Identify the various shapes and arrangements of bacteria in your mouth. Most of them will be bacillus-shaped or coccus-shaped, but it would not be uncommon to see some spirilla. Notice the arrangements of the bacteria---pairs, clusters, chains?
9. Any slide that you use for smears is returned to your slide box, to be cleaned and used again.

Look at a bought prepared bacterial smear

1. Since these are bought stained smears, cover slips are on them. You still use oil on them with the oil-immersion lens.
2. Be sure to REMOVE any oil before replacing them on the trays.

BEFORE PUTTING A SLIDE ON THE MICROSCOPE STAGE

1. Find all of the structures on the microscope ([diagrams at end of exercise](#)) being sure that you know their functions. Rotate the condenser so that you see all of the settings (white letters are engraved into the front of the condenser dial). Also, move the iris diaphragm left and right so you can see the effect on the amount of light.
2. The NIKON microscopes have 3 types of **condensers** for 3 types of light microscopy:
 - o brightfield (O on condenser)
 - o darkfield (DF on condenser)
 - o phase-contrast (condenser settings: PH 1 for 10X, PH3 for 40X, PH4 for 100X)
3. Start with brightfield microscopy **ALWAYS**. The brightfield condenser has a **0** etched in white.
4. Raise the **condenser stage ALL THE WAY UP**. There is a special knob for the condenser stage under the **mechanical stage**. The condenser gathers all available light from the lamp and directs it up to the stage. We always have the condenser stage closest to the mechanical stage when viewing microorganisms. When viewing largest objects, like worms or insects, you can move the condenser down to improve light density hitting the specimen, but not for microorganisms.
5. Turn the **brightness control knob ALL the way up**, and then back off 1/4 of a turn. This is where the control knob will stay: do not touch it again. Your light amount coming up through the condenser is controlled by the **iris diaphragm**.
6. Rotate the **revolving nosepiece** until the **low power 10X objective lens** snaps into place.
7. Bring the stage all the way up, using the **coarse adjustment** knob. Keep an eye on the distance between the slide and the lens to **MAKE SURE** that you do not crash the lens into the stage.
8. Clean all lenses (oculars, objective lenses, and lens on condenser) with **LENS PAPER**.
9. Set the ocular lenses to the correct distance for your face (the oculars can be moved apart or closer together for your own needs). These **ocular eyepiece** lenses are both 10X magnification.



TO VIEW A SPECIMEN

1. Place the wet mount or smear on the stage, and secure it inside of the **stage clips**.
2. Try to guesstimate where the specimen is located on the slide, and place it in the center of the hole allowing light through the stage.
3. While looking through the ocular eyepiece, lower the **stage SLOWLY** using the coarse adjustment knob. Be sure that you are looking through the binocular head of the microscope with **BOTH** eyes.
4. As soon as you see the specimen, **STOP** Using the coarse adjustment, and switch over to the fine adjustment knob. After focusing at the beginning with the coarse adjustment knob, it is **NOT TOUCHED AGAIN**. All focusing will now be done with the **fine adjustment knob**.
5. **CHANGING OBJECTIVES:**

The Nikon lenses are **PARFOCAL**---the objectives are aligned so that rotation to another lens can be done without major focusing. Rotate the 40X lens in place, making sure that it snaps into place. Your specimen should still be seen in the field of vision, but 4 times larger now. Use your fine adjustment knob to clarify the objects.

IF YOUR FIELD OF VISION IS FUZZY, AND NO AMOUNT OF FOCUSING BRINGS THE OBJECT INTO VIEW, YOU PROBABLY HAVE OIL RESIDUE ON THE 40X OBJECTIVE. IT HAS TO BE CLEANED WELL WITH LENS PAPER.



TO MOVE INTO OIL IMMERSION, 100x MAGNIFICATION

1. Do NOT MOVE the focus knobs or the stage knobs. Swing the **40X objective (high dry)** out of the way. Place a single drop of immersion oil on the slide right over where the light is coming through the stage, and rotate the **100X objective (oil immersion)** into place. The lens will actually GO INTO THE OIL DROP.
2. Now look through the oculars, increasing your light with your iris diaphragm lever. Your object should still be in the field of vision, probably out of focus. Use the fine adjustment knob to focus clearly.
3. Once you have gone into oil immersion, do NOT GO BACK TO THE 40X OBJECTIVE. The objective will get oil on it, and you will have to really clean it to get the oil off. The 10X can be returned to, since the lens should not touch the slide anyway.
4. Once through with the microscope, use the *lens paper to wipe the oil* from the 100X objective lens.

USING DARKFIELD AND PHASE-CONTRAST

Once you are looking at your object using brightfield, you can easily switch to another type of microscopy: Just rotate the condenser knob. **HOWEVER:**

- Darkfield is used for wet mounts, using 10X and 40X (100X will not show well). Be sure that your iris diaphragm is OPEN all the way.
- **Phase-contrast** is used for wet mounts also, although SOMETIMES it is helpful for delineating subtle shapes and colors that cannot be readily seen using brightfield. Be sure that you are using the correct condenser setting for that particular objective lens.

PLACING MICROSCOPES BACK INTO THE CABINETS

1. YOU are responsible for your assigned microscope! There is only 1 person in each lab who is assigned that particular microscope, so if someone else complains about it being left with oil or a slide on the stage, you or another person who is assigned that particular scope will be reprimanded.
2. Wrap the cord around the cord holder on the arm.
3. Make sure that the 10X low power lens is in place---not the 100X oil immersion lens. The lens could hit against the stage and get scratched.
4. Turn the coarse adjustment knob so that the stage is far from the lens.
5. PLACE YOUR MICROSCOPE BACK IN ITS NUMBERED POSITION IN THE CORRECT CABINET.

QUESTIONS

1. Which objective lens is also called the oil-immersion lens?
2. How do the functions of the substage condenser and the iris diaphragm within the condenser differ?
3. What term is used to describe the feature of the microscope that makes it possible to move among the objective lenses with just MINOR focusing?
4. What condenser setting value do you want when you are using the 100X objective lens?

Contributors and Attributions

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